

# Proof: Bellman Equations via **do**-Calculus in the MDP Causal Graph

## Abstract

We prove that the Bellman equations for  $v_\pi$  and  $q_\pi$  are identical whether the MDP is interpreted as a Bayesian network (with conditional expectations) or as a causal network (with **do**-interventions). The proof uses Pearl's truncated factorization (g-formula) and the structural properties of the unrolled MDP DAG.

## Contents

### 1 Setup

#### 1.1 The MDP DAG

Consider the unrolled MDP DAG with nodes

$$S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, S_{t+2}, \dots$$

and edges

$$S_t \rightarrow A_t, \quad (S_t, A_t) \rightarrow R_{t+1}, \quad (S_t, A_t) \rightarrow S_{t+1}, \quad S_{t+1} \rightarrow A_{t+1}, \quad \dots$$

The joint distribution factorizes as

$$p(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1}, r_{t+2}, \dots) = p(s_t) \prod_{k \geq 0} \pi(a_{t+k} \mid s_{t+k}) p(r_{t+k+1} \mid s_{t+k}, a_{t+k}, s_{t+k+1}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}).$$

This is both a Bayesian network (the factorization) and a causal network (the edges carry causal meaning).

#### 1.2 The **do**-operator

Pearl's **do**-operator  $\text{do}(X = x)$  means: intervene on  $X$ , set it to  $x$ , and cut all incoming edges to  $X$ . The resulting distribution is given by the truncated factorization (g-formula):

$$p(v_1, \dots, v_n \mid \text{do}(X = x)) = \prod_{i: V_i \neq X} p(v_i \mid \text{pa}(V_i)) \Big|_{X=x}.$$

#### 1.3 Notation

Let

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

be the discounted return from time  $t$ . Define

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid \text{do}(S_t = s)], \quad q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a)].$$

## 2 The truncated factorization for the MDP

**Intervention on  $S_t$  only.** The incoming edges to  $S_t$  come from  $(S_{t-1}, A_{t-1})$ . Cutting them gives

$$\begin{aligned} & p(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1}, r_{t+2}, \dots \mid \text{do}(S_t = s)) \\ &= \mathbf{1}[s_t = s] \pi(a_t \mid s) p(r_{t+1} \mid s, a_t, s_{t+1}) p(s_{t+1} \mid s, a_t) \\ & \quad \times \prod_{k \geq 1} \pi(a_{t+k} \mid s_{t+k}) p(r_{t+k+1} \mid s_{t+k}, a_{t+k}, s_{t+k+1}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}). \end{aligned}$$

**Intervention on both  $S_t$  and  $A_t$ .** The incoming edges to  $A_t$  come from  $S_t$ . Cutting them, and using  $\mathbf{1}[s_t = s]$  and  $\mathbf{1}[a_t = a]$ , gives

$$\begin{aligned} & p(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1}, r_{t+2}, \dots \mid \text{do}(S_t = s), \text{do}(A_t = a)) \\ &= \mathbf{1}[s_t = s] \mathbf{1}[a_t = a] p(r_{t+1} \mid s, a, s_{t+1}) p(s_{t+1} \mid s, a) \\ & \quad \times \prod_{k \geq 1} \pi(a_{t+k} \mid s_{t+k}) p(r_{t+k+1} \mid s_{t+k}, a_{t+k}, s_{t+k+1}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}). \end{aligned}$$

Note that the reward factor  $p(r_{t+1} \mid s, a, s_{t+1})$  depends on  $s_{t+1}$ , which is a variable in the joint. It is not  $p(r_{t+1} \mid s, a)$ , because in the Sutton & Barto convention the reward depends on the transition  $(S_t, A_t, S_{t+1})$ .

## 3 The key observation

In the MDP,  $S_t$  has no parents within the trajectory from time  $t$  onward. The only parents of  $S_t$  are  $(S_{t-1}, A_{t-1})$ , which are past variables. If we restrict attention to the future trajectory and to the return  $G_t$ , then  $S_t$  is a root node in the subgraph of future variables. Therefore,

$$p(\text{future} \mid \text{do}(S_t = s)) = p(\text{future} \mid S_t = s)$$

for the future variables, because  $S_t$  has no parents in the subgraph, and the truncation removes no relevant edges.

For the action-value,  $A_t$  has a parent  $S_t$ . Intervening on  $A_t = a$  cuts the edge  $S_t \rightarrow A_t$ . Since  $S_t$  is fixed at  $s$  in both the intervened and the conditioned distributions, and the edge  $S_t \rightarrow A_t$  is the only path that could cause confusion, the **do**-intervention on  $A_t$  is equivalent to conditioning on  $A_t = a$  given  $S_t = s$ , provided we also intervene on  $S_t = s$ .

## 4 Proof for $v_\pi$

### 4.1 Two expressions to compare

$$\begin{aligned} v_\pi^{\text{Bayes}}(s) &= \mathbb{E}[G_t \mid S_t = s], \\ v_\pi^{\text{causal}}(s) &= \mathbb{E}[G_t \mid \text{do}(S_t = s)]. \end{aligned}$$

### 4.2 The marginal of the reward sequence

**Bayesian case.** Conditioning on  $S_t = s$ ,

$$\begin{aligned} & p(r_{t+1}, r_{t+2}, \dots \mid S_t = s) \\ &= \sum_{a_t, s_{t+1}, a_{t+1}, \dots} \pi(a_t \mid s) p(r_{t+1} \mid s, a_t, s_{t+1}) p(s_{t+1} \mid s, a_t) \pi(a_{t+1} \mid s_{t+1}) \cdots \end{aligned}$$

**Causal case.** By the truncated factorization,

$$\begin{aligned} & p(r_{t+1}, r_{t+2}, \dots \mid \text{do}(S_t = s)) \\ &= \sum_{a_t, s_{t+1}, a_{t+1}, \dots} \pi(a_t \mid s) p(r_{t+1} \mid s, a_t, s_{t+1}) p(s_{t+1} \mid s, a_t) \pi(a_{t+1} \mid s_{t+1}) \cdots \end{aligned}$$

These are identical, because:

- The intervention on  $S_t = s$  cuts the incoming edges to  $S_t$ , which come from  $(S_{t-1}, A_{t-1})$ . These variables do not appear in the sum over future variables.
- The factor for  $S_t$  in the joint becomes  $\mathbf{1}[s_t = s]$ , which is the same as conditioning on  $S_t = s$ .
- All other factors are unchanged.

### 4.3 Conclusion

$$v_\pi^{\text{causal}}(s) = v_\pi^{\text{Bayes}}(s).$$

### 4.4 Derivation of the Bellman equation

Starting from the causal expression:

$$v_\pi(s) = \mathbb{E}[G_t \mid \text{do}(S_t = s)] = \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid \text{do}(S_t = s)].$$

By the truncated factorization,

$$\begin{aligned} & p(a_t, r_{t+1}, s_{t+1}, \text{future} \mid \text{do}(S_t = s)) \\ &= \pi(a_t \mid s) p(r_{t+1} \mid s, a_t, s_{t+1}) p(s_{t+1} \mid s, a_t) p(\text{future} \mid s_{t+1}). \end{aligned}$$

Summing over  $a_t$  and  $s_{t+1}$ :

$$v_\pi(s) = \sum_{a_t, s_{t+1}} \pi(a_t \mid s) p(s_{t+1} \mid s, a_t) [\bar{R}(s, a_t, s_{t+1}) + \gamma v_\pi(s_{t+1})],$$

where  $\bar{R}(s, a, s') = \sum_r r p(r \mid s, a, s')$  and we have used

$$\mathbb{E}[G_{t+1} \mid \text{do}(S_t = s), a_t, s_{t+1}] = \mathbb{E}[G_{t+1} \mid \text{do}(S_{t+1} = s_{t+1})] = v_\pi(s_{t+1}).$$

This is identical to the Bayesian Bellman equation.

## 5 Proof for $q_\pi$

### 5.1 Two expressions to compare

$$\begin{aligned} q_\pi^{\text{Bayes}}(s, a) &= \mathbb{E}[G_t \mid S_t = s, A_t = a], \\ q_\pi^{\text{causal}}(s, a) &= \mathbb{E}[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a)]. \end{aligned}$$

## 5.2 The marginal of the reward sequence

**Bayesian case.** Conditioning on  $S_t = s$  and  $A_t = a$ ,

$$\begin{aligned} & p(r_{t+1}, r_{t+2}, \dots \mid S_t = s, A_t = a) \\ &= \sum_{s_{t+1}, a_{t+1}, \dots} p(r_{t+1} \mid s, a, s_{t+1}) p(s_{t+1} \mid s, a) \pi(a_{t+1} \mid s_{t+1}) \cdots \end{aligned}$$

**Causal case.** By the truncated factorization with interventions on both  $S_t$  and  $A_t$ ,

$$\begin{aligned} & p(r_{t+1}, r_{t+2}, \dots \mid \text{do}(S_t = s), \text{do}(A_t = a)) \\ &= \sum_{s_{t+1}, a_{t+1}, \dots} p(r_{t+1} \mid s, a, s_{t+1}) p(s_{t+1} \mid s, a) \pi(a_{t+1} \mid s_{t+1}) \cdots \end{aligned}$$

These are identical, because:

- The intervention on  $S_t = s$  cuts the incoming edges to  $S_t$ .
- The intervention on  $A_t = a$  cuts the incoming edge to  $A_t$  (from  $S_t$ ). This removes the factor  $\pi(a_t \mid s)$  from the factorization. In the Bayesian case, conditioning on  $A_t = a$  replaces  $\pi(a_t \mid s)$  by  $\mathbf{1}[a_t = a]$ . In both cases,  $A_t$  is fixed at  $a$ .
- All other factors are unchanged.

## 5.3 Conclusion

$$q_\pi^{\text{causal}}(s, a) = q_\pi^{\text{Bayes}}(s, a).$$

## 5.4 Derivation of the Bellman equation

Starting from the causal expression:

$$q_\pi(s, a) = \mathbb{E}[G_t \mid \text{do}(S_t = s), \text{do}(A_t = a)] = \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid \text{do}(S_t = s), \text{do}(A_t = a)].$$

By the truncated factorization,

$$p(r_{t+1}, s_{t+1}, \text{future} \mid \text{do}(S_t = s), \text{do}(A_t = a)) = p(r_{t+1} \mid s, a, s_{t+1}) p(s_{t+1} \mid s, a) p(\text{future} \mid s_{t+1}).$$

Summing over  $s_{t+1}$ :

$$q_\pi(s, a) = \sum_{s_{t+1}} p(s_{t+1} \mid s, a) [\bar{R}(s, a, s_{t+1}) + \gamma v_\pi(s_{t+1})].$$

## 6 Summary

For on-policy evaluation, the Bellman equations for  $v_\pi$  and  $q_\pi$  are identical in the Bayesian and causal interpretations of the MDP. The equivalence relies on the fact that  $S_t$  has no parents in the future subgraph.

*Remark 1.* The corrected factorization for the intervened distribution is

$$\begin{aligned} & p(s_t, a_t, r_{t+1}, s_{t+1}, \dots \mid \text{do}(S_t = s), \text{do}(A_t = a)) \\ &= \mathbf{1}[s_t = s] \mathbf{1}[a_t = a] p(r_{t+1} \mid s, a, s_{t+1}) p(s_{t+1} \mid s, a) \\ &\quad \times \prod_{k \geq 1} \pi(a_{t+k} \mid s_{t+k}) p(r_{t+k+1} \mid s_{t+k}, a_{t+k}, s_{t+k+1}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}). \end{aligned}$$

Note the reward factor depends on  $s_{t+1}$ , not just on  $(s, a)$ .

## Summary of corrections

| Aspect             | Wrong version                              | Corrected version                               |
|--------------------|--------------------------------------------|-------------------------------------------------|
| Indicator function | $\mathbb{I}[s_t = s]$ (needs <b>bold</b> ) | $\mathbf{1}[s_t = s]$ (portable)                |
| Reward factor      | $p(r_{t+1} \mid s, a)$                     | $p(r_{t+1} \mid s, a, s_{t+1})$                 |
| Order of factors   | Product first, then boundary               | Boundary first, then product                    |
| Reward in product  | $p(r_{t+k+1} \mid s_{t+k}, a_{t+k})$       | $p(r_{t+k+1} \mid s_{t+k}, a_{t+k}, s_{t+k+1})$ |